

Robert J. Ramirez Sanchez

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EDUCATION

Ph.D. in Robotics

Mechanical Engineering Department, Virginia Tech, Blacksburg, VA

May 2025 – Present

GPA: 3.92 / 4.00

M.S. in Robotics

Mechanical Engineering Department, Virginia Tech, Blacksburg, VA

May 2025

GPA: 3.92 / 4.00

B.S in Robotics and Mechatronics

Mechanical Engineering Department, Virginia Tech, Blacksburg, VA

May 2024

GPA: 3.94 / 4.00

WORK EXPERIENCE

Graduate Research Assistant

June 2024 - Present

Collaborative Robotics Lab

- Formulated a data augmentation algorithm for learning accelerated policies from human demonstrations.
- Developed a visual imitation learning framework for leveraging task cues from human feedback in the form of natural language and physical markers.
- Developed an algorithm for learning task-relevant representations for efficient policy learning.
- Implemented a representation learning and interface-based algorithm for personalizing robot behaviors.
- Engineered a novel rigid-soft gripper for object manipulation across a 1 million times range in weight.

Mechanical Engineering Co-Op

June 2022 - January 2023

Nutrien

- Designed 3D-printed prototypes for the development of a pipe inspection robot in confined spaces.
- Provided vibration data analysis to support a proposal for pump sled acquisition.
- Designed dragline maintenance platforms and led assembly with engineering and fabrication teams.

PUBLICATIONS

- **R. Ramirez Sanchez**, D. J. Evans, D. P. Losey, and S. Jain, “VOLT: Vision and language trajectory segmentation for faster-than-demonstration policies,” *arXiv preprint arXiv:2606.06323*, 2026, (Under Review).
- Y. Dai*, **R. Ramirez Sanchez***, R. Jeronimus, S. Sagheb, C. M. Nunez, H. Nemlekar, and D. P. Losey, “CIVIL: Causal and intuitive visual imitation learning,” *arXiv preprint arXiv:2504.17959*, 2025, (Under Review).
- **R. Ramirez Sanchez**, H. Nemlekar, S. Sagheb, C. M. Nunez, and D. P. Losey, “RECON: Reducing causal confusion with human-placed markers,” in *IEEE/RSJ International Conference on Intelligent Robots and Systems*, 2025.
- H. Nemlekar, **R. Ramirez Sanchez**, and D. P. Losey, “PECAN: Personalizing robot behaviors through a learned canonical space,” *ACM Transactions on Human-Robot Interaction*, 2025.
- M. Keely, Y. Kim, S. A. Mehta, J. Hoegerman, **R. Ramirez Sanchez**, E. Paul, C. Mills, D. P. Losey, and M. D. Bartlett, “Combining and decoupling rigid and soft grippers to enhance robotic manipulation,” *Soft Robotics*, 2025.

PROJECTS

Automated Binder Jetting De-powdering Using Robot Arms

August 2023 - May 2024

Virginia Tech, sponsored by Azoth3D

- Built a framework for autonomous de-powdering of green parts with motion planners and ROS drivers.
- Automated validation of processed parts through integration of precision measurement sensors.
- Developed microcontroller-based electronic circuits to control a pneumatic system for part processing.

SKILLS

Programming: Python, MATLAB, C/C++

Frameworks: PyTorch, PyBullet, Linux, OpenCV, ROS, Git, Arduino

Robots: Franka Emika Panda, Universal Robots, ABB IRB 1200-7/0.7

3D Modeling: SolidWorks, Autodesk Inventor, Fusion 360